YASIR JAFRI

Senior Software Engineer - Backend & Distributed Systems

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SUMMARY

Senior Software Engineer with 5+ years designing, building, and operating high-traffic, mission-critical backend services on AWS using Python and Go, with full-stack delivery experience shipping Flutter frontends. Sole owner of greenfield platforms serving 1M+ users and 100K+ concurrent entities. Deep experience in event-driven microservices, RESTful API design, real-time stream processing (Kafka, Kafka Streams), and payments/FinTech workflows requiring strict reliability guarantees. Mentors engineers and drives system design, scalability, and reliability decisions.

TECHNICAL SKILLS

Languages: Python (primary), Go, Scala, Java, Node.js (JavaScript), Dart, SQL
Backend & APIs: FastAPI, Django, Flask, Gin, Express.js, Akka, Spark, REST, gRPC, GraphQL, asyncio, WebSockets
Frontend: Flutter (Dart), cross-platform mobile UI, state management, frontend performance optimization, UX flow design
Cloud & DevOps: AWS (Lambda, SNS, SQS, S3, EC2, ECS, EKS, ALB, Auto Scaling, Redshift), Docker, Kubernetes, Jenkins, CI/CD, Git
Data & Messaging: PostgreSQL, MongoDB, Redis, ScyllaDB, ClickHouse, SQL Server, Elasticsearch, Kafka, Kafka Streams
Observability: OpenTelemetry, Honeycomb, SigNoz, distributed tracing, structured logging, metrics, alerting
Architecture: Event-driven microservices, real-time streaming, ETL and medallion data pipelines, system design, multi-tenancy, object-oriented design (SOLID), data structures & algorithms, performance optimization, TDD, DDD, Agile/Scrum
Security: OAuth 2.0, JWT, AES, RBAC, input validation, rate limiting, XSS/CSRF/MITM mitigation, VAPT remediation
AI & GenAI: Agentic AI, LLM integration, prompt engineering, RAG, MCP, AI-assisted SDLC

PROFESSIONAL EXPERIENCE

Senior Software Engineer - Purple Sphere (IDCUBE) - Remote — May 2026 - Present

- Architected a medallion (bronze-silver-gold) data pipeline for a multi-tenant analytics platform supporting unlimited tenant scale at sub-100ms p90 query latency: Kafka ingestion of heterogeneous sources, raw landing in S3, Spark transformations, and Redshift aggregates.
- Built and deployed a modular multi-tenant onboarding platform on AWS EC2 behind an Application Load Balancer with Auto Scaling Group, sustaining sub-200ms p90 API latency under load via RBAC-driven access control and an automated tenant-approval-to-signup workflow with email notifications.
- Standardised engineering workflows across 5 projects and a 4-engineer team by architecting an end-to-end CI/CD pipeline automating linting, formatting, static type checking, testing, build, and deployment, plus structured code reviews that surfaced design issues early.

Senior Software Engineer - Appreciate - Noida — Jul 2024 - Apr 2026

- Revamped the US Stocks module end to end in Flutter, rebuilding the layout and optimising render and data-fetch performance; redesigned the user onboarding experience, lifting onboarding success rate by 20 percent.
- Architected and shipped a real-time stock alert system from scratch as sole owner, tracking 15K-20K symbols for 1M users (about 10K DAU) and delivering about 500 alerts/day at sub-2s end-to-end latency on an event-driven AWS stack (SQS, Lambda, SNS, Redis) with cache-failure fallback and batching.
- Cut critical API latency from 2-3s to under 100ms using asyncio concurrency and distributed tracing with OpenTelemetry on Honeycomb and SigNoz.
- Led the architectural transition from point-to-point API integrations to an SNS-based fan-out model across hundreds of events (checkout, MF/US Stocks onboarding, retention, drop-off tracking), collapsing up to 4 synchronous calls per event into a single publish and reducing new-consumer integration to a config change.
- Built a UPI fund-addition flow from scratch handling about 1,000 transactions/day, eliminating duplicate fund transfers through idempotent order management: payment-status callback polling, internal order-ID deduplication, and 3-retry recovery for transient gateway failures.
- Migrated onboarding flow control from frontend-driven sequencing to a backend FSM-based engine spanning KYC, document verification, and account linking, enabling product-specific onboarding paths, dynamic stage reordering, parallel execution, and support for multiple partner banks.
- Closed all high-priority VAPT findings: input validation, rate limiting, secure HTTP headers, XSS/CSRF and MITM mitigations, and AES encryption for sensitive uploads.
- Mentored a team of up to 8 engineers while embedding Agentic AI into the SDLC through prompt engineering and token optimisation for automated code generation and AI-assisted review and testing, driving 50 percent faster developer iteration cycles and team-wide adoption of SOLID, TDD, and structured review standards.

Software Development Engineer I - Gtropy (MapMyIndia) - Gurugram — Sep 2021 - Jul 2024

- Operated a Kafka and Kafka Streams telemetry pipeline for about 100K vehicles, compressing 20GB to 2GB per cycle via noise filtering, stoppage deduplication, and trip-boundary marking to power trip reconstruction, route optimisation, and fuel/toll analytics.
- Implemented a real-time trip processing engine with concurrent, stateful lifecycle management for 100K+ vehicles using Akka FSM and the Scala actor model, isolating each trip in its own actor so one vehicle's bad data could not disrupt others.
- Migrated performance-critical legacy Django REST APIs to FastAPI, cutting p50 latency from 500ms to 100ms on high-traffic telematics endpoints while remediating vulnerable dependencies; reduced batch GPS cron runtime from 45 minutes to under 3 minutes with asyncio I/O concurrency.
- Led backend for a computer-vision driver-monitoring system across 10-15K vehicles (face recognition, distraction and behaviour detection) at 1-2s detection latency, and built a real-time Twitter scraping and NLP summarisation pipeline feeding multi-tenant B2B client dashboards.

Research Engineer & Technical Faculty - iBotlab - Remote — Jan 2021 - Jun 2021

- Built Python prototypes in data analytics, computer vision, and augmented reality; led a curriculum revamp driving a 30 percent lift in client acquisition.

EDUCATION

M.Tech, Electrical Engineering - Jamia Millia Islamia - GPA 9.43, Gold Medalist — 2018 - 2020

B.Tech, Electrical Engineering - Jamia Millia Islamia - GPA 8.61 — 2013 - 2017